coursera

Gemini Capstone Project ✕

Today's goals
Complete any 3 learning items - 3/3
○ Complete 3 readings - 0/3
Watch a video

Module 1
Function Calling and Agents
✓ Course introduction
Reading • 10 min
✓ From static to dynamic AI
Video • 2 min
✓ Function calling fundamentals
Video • 6 min
✓ Building AI agents
Video • 4 min
✓ Activity: AI agent architecture design
Practice Assignment • Grade: 100%
✓ Evaluating AI agent architecture designs
Dialogue • 15 min
✓ Knowledge check: Building your AI agent
Practice Assignment • Grade: 100%
○ **Conclusion and next steps**
Reading • 10 min

Module 2
Production Planning and Deployment

Conclusion and next steps

0:04 / 10:17 1x

Building intelligent AI agents: From static knowledge to dynamic action

Imagine you have a detailed, printed encyclopedia from a few years ago. This book is incredibly knowledgeable. It contains information on history, science, and culture. If you ask it a question about a historical event, it can give you a perfect, detailed answer based on its fixed information. This is how early AI models worked. They were like a very smart book, filled with a snapshot of the world's information from the time they were created. However, they could not learn anything new about the world as it is right now. If you ask the encyclopedia about today's news, it cannot help you. Its knowledge is static.

Now, think about a modern digital assistant on your phone. It does more than just know facts. It actively connects to the world. It can check today's weather, find the latest news headlines, and look up real-time traffic information. If you ask it to find a nearby coffee shop, it does not just tell you what a coffee shop is; it checks your location and finds open stores near you. This assistant is dynamic; it interacts with a changing environment to help you achieve your goal. This shift from a static book to a dynamic guide is exactly what you have learned to build in this course. You are now equipped with the fundamental skills to transform AI from a simple text generator into an interactive agent that can solve complex, real-world problems.

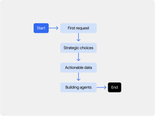
You began with a single, crucial step: making your first request to the Gemini API. You set up your development environment and learned the basic language of AI. You discovered that AI models do not see words the way humans do. Instead, they see "tokens," which are like pieces of words. Understanding tokens helped you see why a long, complex prompt costs more to process than a short one. You also learned about the "context window," which acts as the AI's short-term memory. Everything you send in a request, and every response the AI gives, must fit inside this window. Mastering these core concepts was like learning the alphabet before writing a story. You configured your API key securely, a critical first step for any professional developer. Then, you wrote the code that sent your first prompt and received your first AI-generated response. This was your "Hello, World!" moment in AI development—the foundation upon which every other skill was built.

Once you mastered sending basic requests, you moved from simply using the AI to using it strategically. You then learned how to make smart choices to get the best results. You explored the difference between Gemini models, like Pro and Flash. You learned that choosing a model is a trade-off. Sometimes you need the highest quality and deepest reasoning, like using Gemini Pro to write a detailed technical report. Other times, you need a fast and cost-effective answer, where a model like Flash is the perfect choice. This is like choosing between a detailed, multi-page analysis and a quick, bullet-point summary. You also gained a unique look inside the AI's reasoning process by enabling the "thinking" feature. This tool allowed you to see the steps the model took to arrive at an answer, giving you incredible insight for debugging your prompts and improving the AI's performance. You were no longer just a user of the AI; you became a developer who could guide its behavior and understand its choices.

With the ability to get high-quality responses, the next challenge was making that information useful for other computer systems. An AI's natural language response is easy for a human to read, but it can be difficult for another program to reliably understand . You learned how to solve this problem by getting structured output and JSON Schema to tell the AI exactly what format you wanted the information in. This is the difference between asking the AI to "describe a user" in a long paragraph versus asking it to "fill out this form with the user's name, email, and age." By demanding a predictable structure, you made the AI's output clean, consistent, and ready to be used in a database, an application, or another API. You also explored Gemini's built-in tools. You gave the AI access to Google Search, allowing it to ground its responses in real-time information from the internet. You also used the URL Context tool, which gave the AI the ability to read and analyze the content of a webpage. These tools are like giving the AI pre-made superpowers, instantly expanding what it can do without you having to write a single line of extra code.

This led you to the final part of your journey: teaching the AI to act. The key technology that allows an AI to become a dynamic agent is called function calling. At its core, function calling is a simple but powerful idea: it gives the AI a way to pause, ask your code to perform a task, and then use the result of that task to give a better answer. Think of it as giving the AI a phone. If the AI does not know something or cannot do something on its own, it can "call" a program you have written to get help. This is the bridge that connects the AI's digital brain to the real world.

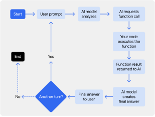
The modern AI developer's journey



Let's look at a practical example. Imagine a customer is talking to an AI chatbot on a retail website. The customer asks, "Are the blue running shoes in stock in a size 10?" A static AI would fail. But a dynamic AI, equipped with function calling, understands that it needs information from the company's inventory system. It calls a special function the developer created, perhaps named ` check_inventory` . Your code receives this request, connects to the company's database, checks the stock level, and sends the answer back to the AI. Now, with this new, real-time information, the AI can give a helpful response: "Yes, we have three pairs left in stock. Would you like me to add a pair to your cart?"

This interaction happens in a two-step process, often called a two-turn loop. First, the AI analyzes the user's prompt and sends a request to your code to run a function. Second, your code runs the function and sends the result back to the AI, which then creates the final response for the user. This system makes the AI a manager or orchestrator. It decides what needs to be done, and your code is the skilled employee that performs the work.

The two-turn function calling loop


Start
User prompt
AI model analyzes
AI requests function call
Your code executes the function
Function result returned to AI
AI model creates final answer
Final answer to user
Another turn?
Yes
No
End

To make this work, you learned that you must provide the AI with a clear description of each tool in your toolkit. The AI relies completely on your natural language description to understand what a function does and when to use it. As an AI architect, your job is to build a complete toolkit for your agent, combining custom functions with built-in tools. For example, as a travel agent, you might use the built-in Google Search to find current events but a custom function to connect to a specific airline's booking system. By building small, single-purpose tools, you give the AI the flexibility to combine them in new and creative ways to solve user problems.

In this course, you have assembled all the pieces needed to build a complete, intelligent agent. A great agent is not about a single skill; it is the combination of everything you have learned. It starts with a well-formed API call. It uses the right model for the task. It gets real-time information using built-in tools and provides reliable, structured output. And it connects to the real world to perform actions using custom functions.

Your role as a developer has evolved. You are no longer just writing instructions. You are an architect designing systems. You create high-quality, reliable tools, and the AI uses its reasoning ability to orchestrate those tools to solve problems. As you build these production-ready systems, you must remember the best practices you have learned. Always secure your functions by validating inputs. Always handle errors gracefully, so your application does not crash. And always consider performance by using techniques like asynchronous execution to keep your application responsive. With these skills, you are now at the forefront of modern application development, ready to build the next generation of intelligent, helpful, and dynamic AI agents.

Coming up next: Exploring Google AI Studio

You have built an incredible foundation by learning to code with the Gemini API. Your next step is to accelerate your workflow. In the next course on Google AI Studio, you will learn about rapid prototyping and experimentation. You will learn to fine-tune AI outputs by adjusting settings, visualize token usage to make cost-effective decisions, and swiftly develop and test new application ideas in a hands-on, interactive environment.

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